

|    |     |       |                   |                                                            |                                     |                                                      |
|----|-----|-------|-------------------|------------------------------------------------------------|-------------------------------------|------------------------------------------------------|
|    |     |       |                   |                                                            |                                     |                                                      |
| 16 | B   |       |                   |                                                            |                                     |                                                      |
|    |     | Solid | Melting point/ °C | Specific heat capacity/ J kg <sup>-1</sup> K <sup>-1</sup> | Δ <i>T</i> from 20 °C to melting pt | Energy required for 1 kg to reach melting point / kJ |
|    |     | A     | 80                | 1200                                                       | 60                                  | 72                                                   |
|    |     | B     | 100               | 800                                                        | 80                                  | 64                                                   |
|    |     | C     | 150               | 600                                                        | 130                                 | 78                                                   |
| D  | 300 | 250   | 280               | 70                                                         |                                     |                                                      |

|    |   |                                                                                                                                                                                                                            |
|----|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    |   | Based on the energy required for each solid to reach their melting point, solid B which requires the least amount of energy to reach its melting point will melt first since they are all heated at same rate ( $E = Pt$ ) |
| 17 | C | With partial vacuum, the max amplitude would be larger and natural frequency                                                                                                                                               |